

Workload-Aware Hybrid Policy

Prefill-heavy

Attention q/k/v/o

2:4 BF16

MLP expand gate/up

2:4 W4A4

MLP reduce down

2:4 BF16

Balanced prefill + decode

All language linear layers

W4A4

->

W4A16

Decode-heavy

All language linear layers

W4A4

->

W4A16

Hybrid selection is workload-aware, not one-size-fits-all

■ 2:4 BF16

■ 2:4 W4A4

■ Hybrid: W4A4 -> W4A16